

Honours Multivariate Analysis

Lecture 3 - The Multivariate Normal Distribution

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Course Outline

- ① Introduction, Examples of Multivariate Data [†]
- ② TOOLS
 - ① Visualization and Summary Statistics [†]
 - ② Singular Value Decomposition, Eigenvalue Decomposition and Spectral Decomposition revisited [†]
 - ③ **The Multivariate Normal Distribution** [†]
 - ④ Multivariate Maximum Likelihood Estimation [†]
 - ⑤ Multivariate Inference [†]
- ③ EXPLORATORY ANALYSIS
 - ① Principal Component Analysis [§]
 - ② Factor Analysis [§]
 - ③ Correspondence Analysis [§]
- ④ CONFIRMATORY ANALYSIS
 - ① For grouped Multivariate Data:
 - ① Manova [†]
 - ② Discriminant Analysis [§]
 - ② Regression
 - ① Multivariate Regression [§]
 - ② Canonical Correlation Analysis [§]

[†]Mr Stefan Britz

[§]Mr Miguel Rodó

The Univariate Normal Distribution

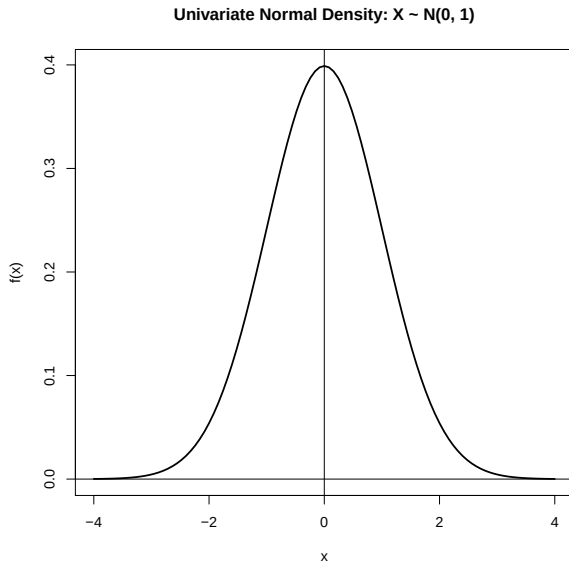
The univariate normal probability density function for a random variable $X \sim N(\mu, \sigma^2)$ is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

where

- $\frac{1}{\sigma\sqrt{2\pi}}$ is a normalizing constant
- $\frac{(x-\mu)^2}{\sigma^2} = \underbrace{(x-\mu)\sigma^{-2}(x-\mu)}_{\text{square of distance from } x \text{ to } \mu \text{ in std.dev. units}}$

The Univariate Normal Distribution



The Multivariate Normal Distribution

Consider now $\mathbf{X}_{p \times 1}$, with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. The multivariate generalized distance is

$$(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

and the multivariate version of the normalizing constant is

$$(2\pi)^{-\frac{p}{2}} |\boldsymbol{\Sigma}|^{-\frac{1}{2}}.$$

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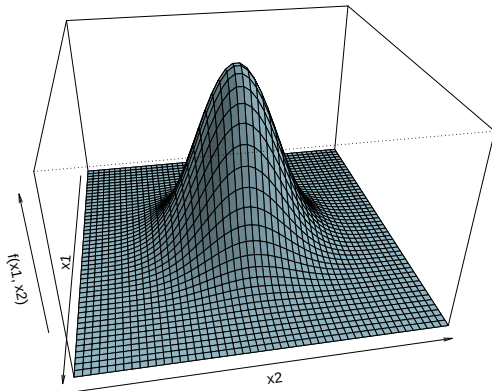
Thus the multivariate normal density function is

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{p}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right], \quad \forall \mathbf{x}$$

and we write $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

The Bivariate Normal Distribution

When $p = 2$, we have the bivariate normal distribution.



See `Lecture3.R` for more examples of plots, and [this interactive illustration](#).

The Bivariate Normal Distribution

When $p = 2$,

- $\mu_1 = E(X_1)$, $\mu_2 = E(X_2)$,
- $\sigma_{11} = \text{Var}(X_1)$, $\sigma_{22} = \text{Var}(X_2)$, $\sigma_{12} = \sigma_{21} = \text{Cov}(X_1, X_2)$
- $\rho_{12} = \frac{\sigma_{12}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{22}}} = \text{Cor}(X_1, X_2)$

Using $\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix}$, we can calculate

$$\Sigma^{-1} = \frac{1}{\sigma_{11}\sigma_{22} - \sigma_{12}^2} \begin{bmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{12} & \sigma_{11} \end{bmatrix} = \frac{1}{\sigma_{11}\sigma_{22}(1 - \rho_{12}^2)} \begin{bmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{12} & \sigma_{11} \end{bmatrix},$$

since $\sigma_{12}^2 = \sigma_{11}\sigma_{22}\rho_{12}^2$. Therefore, we have

The Bivariate Normal Distribution

$$\begin{aligned} & (\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\ &= \frac{1}{\sigma_{11}\sigma_{22}(1 - \rho_{12}^2)} \times \\ & \begin{bmatrix} x_1 - \mu_1 & x_2 - \mu_2 \end{bmatrix} \begin{bmatrix} \frac{\sigma_{22}}{\sigma_{11}\sqrt{1 - \rho_{12}^2}} & -\frac{\rho_{12}\sqrt{\sigma_{22}}}{\sigma_{11}\sqrt{1 - \rho_{12}^2}} \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix} \\ &= \frac{\sigma_{22}(x_1 - \mu_1)^2 + \sigma_{11}(x_2 - \mu_2)^2 - 2\rho_{12}\sqrt{\sigma_{11}\sigma_{22}}(x_1 - \mu_1)(x_2 - \mu_2)}{\sigma_{11}\sigma_{22}(1 - \rho_{12}^2)} \\ &= \frac{1}{1 - \rho_{12}^2} \times \left[\left(\frac{x_1 - \mu_1}{\sqrt{\sigma_{11}}} \right)^2 + \left(\frac{x_2 - \mu_2}{\sqrt{\sigma_{22}}} \right)^2 - 2\rho_{12} \left(\frac{x_1 - \mu_1}{\sqrt{\sigma_{11}}} \right) \left(\frac{x_2 - \mu_2}{\sqrt{\sigma_{22}}} \right) \right] \end{aligned}$$

The Bivariate Normal Distribution

We can write the normalizing constant as

$$\frac{1}{(2\pi)^{\frac{p}{2}}|\mathbf{\Sigma}|^{1/2}} = \frac{1}{2\pi\sqrt{|\mathbf{\Sigma}|}} = \frac{1}{2\pi\sqrt{\sigma_{11}\sigma_{22}(1-\rho_{12}^2)}}$$

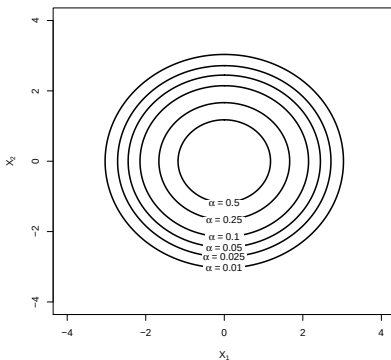
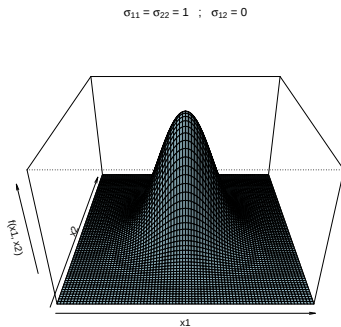
Thus

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_{11}\sigma_{22}(1-\rho_{12}^2)}} \times e^{-\frac{1}{2(1-\rho_{12}^2)} \times \left[\left(\frac{x_1-\mu_1}{\sqrt{\sigma_{11}}} \right)^2 + \left(\frac{x_2-\mu_2}{\sqrt{\sigma_{22}}} \right)^2 - 2\rho_{12} \left(\frac{x_1-\mu_1}{\sqrt{\sigma_{11}}} \right) \left(\frac{x_2-\mu_2}{\sqrt{\sigma_{22}}} \right) \right]}$$

If X_1 and X_2 are uncorrelated, $\rho_{12} = 0$ and $f(x_1, x_2) = f(x_1) \times f(x_2)$.

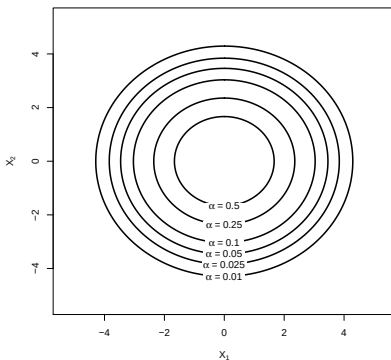
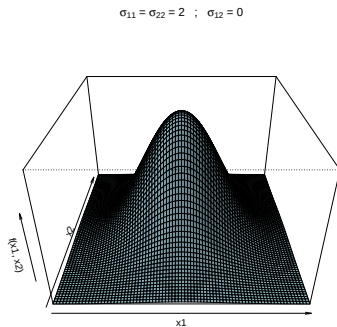
Bivariate Contours

The shape of a bivariate distribution and the relationship between the variables can also be explored via a contour plot.



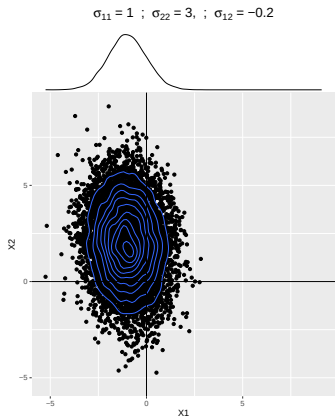
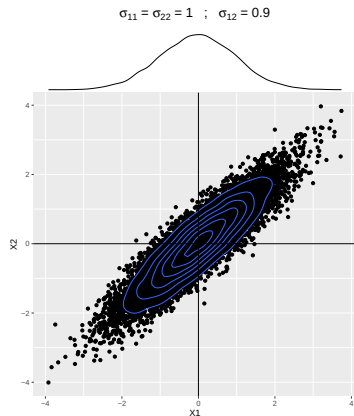
Bivariate Contours

The shape of a bivariate distribution and the relationship between the variables can also be explored via a contour plot.



Bivariate Contours

We can also draw contours around data simulated from bivariate normal distributions.



Again, see `Lecture3.R`.

Homework exercise 3.1

Johnson & Wichern exercise 4.2

Consider a bivariate normal population with

$$\mu_1 = 0, \mu_2 = 2, \sigma_{11} = 2, \sigma_{22} = 1, \rho_{12} = 0.5$$

- 1 Write out the bivariate normal density.
- 2 Write out the squared generalized distance expression $(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$ as a function of x_1 and x_2 .

Graphical interpretation of bivariate normal density

- The multivariate normal density is constant on surfaces where the square of the distance from x to the mean, μ is constant.

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- These constant probability contours are defined as all $\mathbf{x}$ such that $(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2$.

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- These constant probability contours are defined as all $\mathbf{x}$ such that $(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2$.
- The axes of each ellipsoid of constant density are in the direction of the eigenvectors of $\boldsymbol{\Sigma}^{-1}$ and their lengths are proportional to the reciprocals of the square roots of the eigenvalues of $\boldsymbol{\Sigma}^{-1}$.
- Therefore, the axes are $\pm c \sqrt{\lambda_i} \mathbf{e}_i$, where the eigendecomposition of $\boldsymbol{\Sigma}$ is $\boldsymbol{\Sigma} \mathbf{e}_i = \lambda_i \mathbf{e}_i$ for $i = 1, 2, \dots, p$.

Bivariate normal density example

Assume $\sigma_{11} = \sigma_{22}$. Thus the characteristic equation $|\Sigma - \lambda I| = 0$ becomes

$$\begin{aligned} 0 &= \begin{vmatrix} \sigma_{11} - \lambda & \sigma_{12} \\ \sigma_{12} & \sigma_{11} - \lambda \end{vmatrix} \\ &= (\sigma_{11} - \lambda)^2 - \sigma_{12}^2 \\ &= (\lambda - \sigma_{11} - \sigma_{12})(\lambda - \sigma_{11} + \sigma_{12}) \end{aligned}$$

$\therefore$ the eigenvalues are $\lambda_1 = \sigma_{11} + \sigma_{12}$ and $\lambda_2 = \sigma_{11} - \sigma_{12}$.

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The first eigenvector, e_1 , is then determined from

$$\begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{11} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = (\sigma_{11} + \sigma_{12}) \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

which gives us

$$\sigma_{11}e_1 + \sigma_{12}e_2 = (\sigma_{11} + \sigma_{12})e_1$$

$$\sigma_{12}e_1 + \sigma_{11}e_2 = (\sigma_{11} + \sigma_{12})e_2$$

$$\implies e_1 = e_2.$$

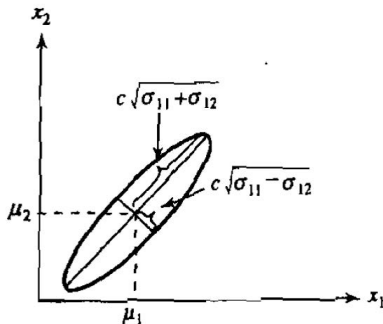
Bivariate normal density example

After normalization we have that the first eigenvalue-eigenvector pair is

$$\lambda_1 = \sigma_{11} + \sigma_{12}, \quad e_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}'$$

Likewise, the second pair is

$$\lambda_2 = \sigma_{11} - \sigma_{12}, \quad e_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}'$$



The Probability Content of the Ellipsoids of Constant Density

Let $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $|\boldsymbol{\Sigma}| > 0$. Then

- ① $(\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu})$ is distributed as χ_p^2 .
- ② The $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution assigns a probability of $1 - \alpha$ to the solid ellipsoid

$$\{\mathbf{x} : (\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \leq \chi_p^2(\alpha)\},$$

where $\chi_p^2(\alpha)$ denotes the upper (100α) th percentile of the χ_p^2 distribution.

That is,

$$\Pr [(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \leq \chi_p^2(\alpha)] = 1 - \alpha.$$

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Homework exercise 3.2

- 1 Prove the above by using the spectral decomposition of the covariance matrix.
- 2 Determine (and sketch) the constant-density contour that contains 90% of the probability for the examples in exercise 3.1.

Properties of the Multivariate Normal Distribution 1

Linear combinations of the components of $\mathbf{X}$ are normally distributed.

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then ANY linear combination

$$\mathbf{a}'\mathbf{X} = a_1X_1 + a_2X_2 + \dots + a_pX_p \sim N(\mathbf{a}'\boldsymbol{\mu}, \mathbf{a}'\boldsymbol{\Sigma}\mathbf{a})$$

and vice-versa, if $\mathbf{a}'\mathbf{X} \sim N(\mathbf{a}'\boldsymbol{\mu}, \mathbf{a}'\boldsymbol{\Sigma}\mathbf{a})$ for every $\mathbf{a}$, then $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Also, $\mathbf{X} + \mathbf{d} \sim N_p(\boldsymbol{\mu} + \mathbf{d}, \boldsymbol{\Sigma})$ where $\mathbf{d}$ is a vector of constants.

Linear combinations of the components of $\mathbf{X}$ are normally distributed

Example:

Let $\mathbf{a}' = [1, 0, \dots, 0]$, then

$$\mathbf{a}'\mathbf{X} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix} = X_1$$

$$\mathbf{a}'\boldsymbol{\mu} = \mu_1$$

$$\mathbf{a}'\boldsymbol{\Sigma}\mathbf{a} = \sigma_{11}$$

$$\rightarrow \mathbf{a}'\mathbf{X} \sim N(\mu_1, \sigma_{11})$$

Homework exercise 3.3

Johnson & Wichern exercise 4.4(a)

Given $\mathbf{X} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu}' = [2, -3, 1]$ and $\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 2 \end{bmatrix}$,

find the distribution of $3X_1 - 2X_2 + X_3$.

Properties of the Multivariate Normal Distribution 2

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then q linear combinations, then

$$\underbrace{\mathbf{A}}_{q \times p} \underbrace{\mathbf{X}}_{p \times 1} = \begin{bmatrix} a_{11}X_1 + \dots + a_{1p}X_p \\ a_{21}X_1 + \dots + a_{2p}X_p \\ \vdots \\ a_{q1}X_1 + \dots + a_{qp}X_p \end{bmatrix} \sim N_q(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$$

q linear combinations of $\mathbf{X}$

Example:

If $\mathbf{X} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$, then

$$\mathbf{A}\mathbf{X} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} X_1 - X_2 \\ X_2 - X_3 \end{bmatrix}$$

$$\mathbf{A}\boldsymbol{\mu} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} = \begin{bmatrix} \mu_1 - \mu_2 \\ \mu_2 - \mu_3 \end{bmatrix}$$

$$\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}' = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}$$

Properties of the Multivariate Normal Distribution 3

All subsets of the components of $\mathbf{X}$ have a (multivariate) normal distribution.
If we partition $\mathbf{X}$ as

$$\mathbf{X}_{p \times 1} = \begin{bmatrix} \underbrace{\mathbf{X}_1}_{q \times 1} \\ \underbrace{\mathbf{X}_2}_{(p-q) \times 1} \end{bmatrix}$$

then

$$\boldsymbol{\mu}_{p \times 1} = \begin{bmatrix} \underbrace{\boldsymbol{\mu}_1}_{q \times 1} \\ \underbrace{\boldsymbol{\mu}_2}_{(p-q) \times 1} \end{bmatrix} \quad \text{and} \quad \boldsymbol{\Sigma}_{p \times p} = \begin{bmatrix} \underbrace{\boldsymbol{\Sigma}_{11}}_{q \times q} & \underbrace{\boldsymbol{\Sigma}_{12}}_{q \times (p-q)} \\ \underbrace{\boldsymbol{\Sigma}_{21}}_{(p-q) \times q} & \underbrace{\boldsymbol{\Sigma}_{22}}_{(p-q) \times (p-q)} \end{bmatrix}$$

yielding $\mathbf{X}_1 \sim N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $\mathbf{X}_2 \sim N_{(p-q)}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$.

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yielding $\mathbf{X}_1 \sim N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $\mathbf{X}_2 \sim N_{(p-q)}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$.

The result follows from defining $\mathbf{A} = \begin{bmatrix} \underbrace{\mathbf{I}}_{q \times q} & \underbrace{\mathbf{0}}_{q \times (p-q)} \end{bmatrix}$ and applying property 2.

Properties of the Multivariate Normal Distribution 4

Zero covariance implies that the corresponding components are independently distributed.

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1 If $\underbrace{\mathbf{X}_1}_{q_1 \times 1}$ and $\underbrace{\mathbf{X}_2}_{q_2 \times 1}$ are independent, then $\text{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \underbrace{\mathbf{0}}_{q_1 \times q_2}$.

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- 2 If $\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} \sim N_{q_1+q_2} \left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \right)$ then $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent if and only if $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.

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- 1 If $\underbrace{\mathbf{X}_1}_{q_1 \times 1}$ and $\underbrace{\mathbf{X}_2}_{q_2 \times 1}$ are independent, then $\text{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \underbrace{\mathbf{0}}_{q_1 \times q_2}$.
- 2 If $\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} \sim N_{q_1+q_2} \left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \right)$ then $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent if and only if $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.
- 3 If $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent and are distributed as $N_{q_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $N_{q_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$, respectively, then $\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} \sim N_{q_1+q_2} \left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \right)$.

Zero covariance implies that the corresponding components are independently distributed

Example:

If we let $\underbrace{\mathbf{X}}_{3 \times 1} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, then X_1 and X_2 are NOT independent, since $\sigma_{12} = 1$.

BUT if we partition $\mathbf{X}$ as follows:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \left[\begin{array}{cc|c} 4 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{array} \right]$$

then

$\mathbf{X}_1 = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ and X_3 have covariance matrix $\boldsymbol{\Sigma}_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and so $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ and X_3 are independent.

Homework exercise 3.4

Johnson & Wichern exercise 4.3

Let $\mathbf{X} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu}' = [-3 \quad 1 \quad 4]$ and $\boldsymbol{\Sigma} = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Which of the following pairs of random variables are independent? Explain.

- ① X_1 and X_2
- ② X_2 and X_3
- ③ (X_1, X_2) and X_3
- ④ $\frac{X_1 + X_2}{2}$ and X_3

Properties of the Multivariate Normal Distribution 5

The conditional distributions of the components are (multivariate) normally distributed

$$\text{If } \mathbf{X} = \begin{bmatrix} \underbrace{\mathbf{X}_1}_{q \times 1} \\ \underbrace{\mathbf{X}_2}_{(p-q) \times 1} \end{bmatrix} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \text{ with } \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \text{ and } \boldsymbol{\Sigma} = \left[\begin{array}{c|c} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \hline \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{array} \right]$$

then the conditional distribution of $\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2$ is multivariate normal with mean

$$E(\mathbf{X}_1 | \mathbf{X}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{X}_2 - \boldsymbol{\mu}_2)$$

and covariance matrix

$$\text{Cov}(\mathbf{X}_1 | \mathbf{X}_2) = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}.$$

Properties of the Multivariate Normal Distribution 5

The conditional distributions of the components are (multivariate) normally distributed

$$\text{If } \mathbf{X} = \begin{bmatrix} \underbrace{\mathbf{X}_1}_{q \times 1} \\ \underbrace{\mathbf{X}_2}_{(p-q) \times 1} \end{bmatrix} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \text{ with } \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \text{ and } \boldsymbol{\Sigma} = \left[\begin{array}{c|c} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \hline \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{array} \right]$$

then the conditional distribution of $\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2$ is multivariate normal with mean

$$E(\mathbf{X}_1 | \mathbf{X}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{X}_2 - \boldsymbol{\mu}_2)$$

and covariance matrix

$$\text{Cov}(\mathbf{X}_1 | \mathbf{X}_2) = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}.$$

Make sure that you go through the proof for this provided in the notes!

Homework exercise 3.5

Johnson & Wichern 4.5(a)

Consider again the bivariate normal population with $\mu_1 = 0, \mu_2 = 2, \sigma_{11} = 2, \sigma_{22} = 1, \rho_{12} = 0.5$.

Specify the conditional distribution of $X_1|X_2 = x_2$.

Summary of property 5

① All conditional distributions are multivariate normal.

② The conditional mean is of the form

$$\begin{aligned} \mu_1 + \beta_{1,q+1}(x_{q+1} - \mu_{q+1}) + \dots + \beta_{1,p}(x_p - \mu_p) \\ \vdots \\ \mu_q + \beta_{q,q+1}(x_{q+1} - \mu_{q+1}) + \dots + \beta_{q,p}(x_p - \mu_p) \end{aligned},$$

where the β' s are defined by

$$\Sigma_{12}\Sigma_{22}^{-1} = \begin{bmatrix} \beta_{1,q+1} & \beta_{1,q+2} & \dots & \beta_{1,p} \\ \beta_{2,q+1} & \beta_{2,q+2} & \dots & \beta_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{q,q+1} & \beta_{q,q+2} & \dots & \beta_{q,p} \end{bmatrix}$$

③ The conditional covariance $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$ does not depend upon the values of the conditioning variables.

Linear combinations of random variable vectors

$$\text{Consider } \underbrace{\mathbf{V}_1}_{p \times 1} = c_1 \mathbf{X}_1 + c_2 \mathbf{X}_2 + \dots + c_n \mathbf{X}_n = \underbrace{[\mathbf{X}_1 \quad \mathbf{X}_2 \quad \dots \quad \mathbf{X}_n]}_{(p \times n)} \underbrace{\mathbf{c}}_{n \times 1}$$

where the $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are mutually independent with each $\mathbf{X}_j \sim N_p(\boldsymbol{\mu}_j, \boldsymbol{\Sigma})$.

$$\text{Then } \mathbf{V}_1 \sim N_p \left(\sum_{j=1}^n c_j \boldsymbol{\mu}_j, \left(\sum_{j=1}^n c_j^2 \right) \boldsymbol{\Sigma} \right).$$

Linear combinations of random variable vectors

$$\text{Consider } \underbrace{\mathbf{V}_1}_{p \times 1} = c_1 \mathbf{X}_1 + c_2 \mathbf{X}_2 + \dots + c_n \mathbf{X}_n = \underbrace{\begin{bmatrix} \mathbf{X}_1 & \mathbf{X}_2 & \dots & \mathbf{X}_n \end{bmatrix}}_{(p \times n)} \underbrace{\mathbf{c}}_{n \times 1}$$

where the $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are mutually independent with each $\mathbf{X}_j \sim N_p(\boldsymbol{\mu}_j, \boldsymbol{\Sigma})$.

Then $\mathbf{V}_1$ and $\mathbf{V}_2 = b_1 \mathbf{X}_1 + b_2 \mathbf{X}_2 + \dots + b_n \mathbf{X}_n$ are jointly multivariate normal with covariance matrix

$$\begin{bmatrix} \left(\sum_{j=1}^n c_j^2 \right) \boldsymbol{\Sigma} & (\mathbf{b}' \mathbf{c}) \boldsymbol{\Sigma} \\ (\mathbf{b}' \mathbf{c}) \boldsymbol{\Sigma} & \left(\sum_{j=1}^n b_j^2 \right) \boldsymbol{\Sigma} \end{bmatrix}$$

So $\mathbf{V}_1$ and $\mathbf{V}_2$ are independent if $\mathbf{b}' \mathbf{c} = \sum_{j=1}^n c_j b_j = 0$.

Homework exercise 3.6

Johnson & Wichern exercises 4.16 and 4.17

Let $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3$ and $\mathbf{X}_4$ be independent $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ random vectors.

- ① Find the marginal distributions of

$$\mathbf{V}_1 = \frac{1}{4}\mathbf{X}_1 - \frac{1}{4}\mathbf{X}_2 + \frac{1}{4}\mathbf{X}_3 - \frac{1}{4}\mathbf{X}_4$$

and

$$\mathbf{V}_2 = \frac{1}{4}\mathbf{X}_1 + \frac{1}{4}\mathbf{X}_2 - \frac{1}{4}\mathbf{X}_3 - \frac{1}{4}\mathbf{X}_4$$

- ② Find the joint density of $\mathbf{V}_1$ and $\mathbf{V}_2$.

Homework exercise 3.6

Let $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_4$ and $\mathbf{X}_5$ be independent and identically distributed random vectors with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$.

- ④ Find the mean vector and covariance matrices for

$$\frac{1}{5}\mathbf{X}_1 + \frac{1}{5}\mathbf{X}_2 + \frac{1}{5}\mathbf{X}_3 + \frac{1}{5}\mathbf{X}_4 + \frac{1}{5}\mathbf{X}_5$$

and

$$\mathbf{X}_1 - \mathbf{X}_2 + \mathbf{X}_3 - \mathbf{X}_4 + \mathbf{X}_5$$

in terms of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. Also, obtain the covariance between these two linear combinations.